



Lab Report - 05

Experiment on For loop in C programming

Course Name: Structured Programming Lab

Course Code: CSE-121

Submitted to

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Submission Date: 2025-07-26

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05	Write a program in C to display the cube of the number up to an integer.
06	Write a program in C to display the multiplication table for a given integer.
07	Write a C program to display the n terms of odd natural numbers and their sum.

Signature:

Date:

Problem-1:Write a C program to print your name ten times.

ExpectedOutput:

Name Name Name Name Name Name Name Name Name Name

Source Code:

```
*Untitled1.c X
1  #include<stdio.h>
2  int main()
3  {
4      int i;
5      for(i=1;i<=10;i++)
6          { printf("Anika ");}
7
8      return 0;
9  }
```

Output:

```
Anika Anika Anika Anika Anika Anika Anika Anika Anika Anika
Process returned 0 (0x0)   execution time : 0.156 s
Press any key to continue.
```

Problem-2: Write a program in C to print numbers from 0 to 10 and 10 to 0 using loops.

Expected Output :

Numbers from 0 to 10:

0 1 2 3 4 5 6 7 8 9 10

Numbers from 10 to 0:

10 9 8 7 6 5 4 3 2 1 0

Source Code:

```
e X Untitled1.c X
1  #include<stdio.h>
2  int main()
3  {
4      int i;
5
6      printf("Numbers from 0 to 10:\n");
7      for(i=0; i<=10; i++)
8          {printf("%d ",i);}
9
10     printf("\n\nNumbers from 10 to 0:\n");
11     for(i=10; i>=0; i--)
12         { printf("%d ",i);}
13     printf("\n");
14
15     return 0;
16 }
17
```

Output:

```
Numbers from 0 to 10:
0 1 2 3 4 5 6 7 8 9 10

Numbers from 10 to 0:
10 9 8 7 6 5 4 3 2 1 0

Process returned 0 (0x0)    execution time : 0.160 s
Press any key to continue.
|
```

Problem-3: Write a C program to compute the sum of the first 10 natural numbers.

Expected Output:

The first 10 natural numbers are:

1 2 3 4 5 6 7 8 9 10

The Sum is: 55

Source Code:

```
ere X *Untitled1.c X
1  #include<stdio.h>
2  int main()
3  {
4      int sum,i;
5      printf("The first 10 natural numbers are:\n");
6      for( i=1;i<=10;i++)
7          { printf("%d ",i);
8            sum=sum+i;}
9
10     printf("\nThe Sum is:%d\n",sum);
11
12     return 0;
13 }
14
```

Output:

```
The first 10 natural numbers are:
1 2 3 4 5 6 7 8 9 10
The Sum is:55

Process returned 0 (0x0)   execution time : 0.139 s
Press any key to continue.
```

Problem-4: Write a program in C to read 10 numbers from the keyboard and find their sum and average.

Test Data :

Input the 10 numbers :

Number-1 :2

...

Number-10 :2

Expected Output:

The sum of 10 no is: 55

The Average is: 5.500000

Source Code:

```
ere X  Untitled4.c X
1  #include<stdio.h>
2  int main()
3  {
4      int num,i,sum=0; float avg;
5      printf("Input the 10 numbers:\n");
6      for( i=1; i<=10 ; i++)
7      {printf("Number-%d:",i);
8          scanf("%d",&num);
9          sum=sum+num;}
10     printf("\nThe sum of 10 no is:%d",sum);
11     avg=sum/10.0;
12     printf("\nThe Average is:%f\n",avg);
13
14     return 0;
15 }
16
```

Output:

```
Input the 10 numbers:
Number-1:2
Number-2:3
Number-3:4
Number-4:5
Number-5:6
Number-6:7
Number-7:8
Number-8:9
Number-9:2
Number-10:4

The sum of 10 no is:50
The Average is:5.000000

Process returned 0 (0x0)   execution time : 16.520 s
Press any key to continue.
|
```

Problem-5: Write a program in C to display the cube of the number up to an integer.

Test Data :

Input number of terms: 5

Expected Output:

Number is : 1 and cube of the 1 is :1

Number is : 2 and cube of the 2 is :8

Number is : 3 and cube of the 3 is :27

Number is : 4 and cube of the 4 is :64

Number is : 5 and cube of the 5 is :125

Source Code:

```
e X Untitled7.c X
1  #include<stdio.h>
2  int main()
3  {
4      int num,i;
5      printf("Input number of terms:");
6      scanf("%d",&num);
7
8      for(i=1; i<=num ;i++)
9      {
10         printf("Number is:%d and cube of the %d is:%d\n",i,i,i*i*i);
11     }
12     return 0;
13 }
```

Output:

```
Input number of terms:5
Number is:1 and cube of the 1 is:1
Number is:2 and cube of the 2 is:8
Number is:3 and cube of the 3 is:27
Number is:4 and cube of the 4 is:64
Number is:5 and cube of the 5 is:125

Process returned 0 (0x0)   execution time : 2.109 s
Press any key to continue.
|
```


Problem-6: Write a program in C to display the multiplication table for a given integer.

Test Data :

Input the number (Table to be calculated): 15

Expected Output:

15 X 1 = 15

...

...

15 X 10 = 150

Source Code:

```
e x Untitled8.c x
1  #include<stdio.h>
2  int main()
3  {
4      int num,i,multi;
5      printf("Input the number (Table to be calculated):");
6      scanf("%d",&num);
7
8      for( i=1; i<=10; i++)
9      {multi=num*i;
10         printf("%d X %d=%d\n",num,i,multi);}
11
12     return 0;
13 }
14
```

Output:

```
Input the number (Table to be calculated):15
15 X 1=15
15 X 2=30
15 X 3=45
15 X 4=60
15 X 5=75
15 X 6=90
15 X 7=105
15 X 8=120
15 X 9=135
15 X 10=150

Process returned 0 (0x0)   execution time : 2.972 s
Press any key to continue.
```

Problem-7: Write a C program to display the n terms of odd natural numbers and their sum.

Test Data :

Input number of terms: 10

Expected Output:

The odd numbers are :1 3 5 7 9 11 13 15 17 19

The Sum of odd Natural Numbers up to 10 terms: 100

Source Code:

```
1  #include<stdio.h>
2  int main()
3  {
4      int num,i,sum=0,count=0;
5      printf("Input numbers of terms:");
6      scanf("%d",&num);
7
8      printf("The odd numbers are:");
9      for(i=1; count<num; i=i+2)
10     {if(i%2!=0)
11         printf("%d ",i);
12         sum=sum+i;
13         count++;}
14     printf("\nThe Sum of odd Natural up to 10 terms:%d\n",sum);
15
16     return 0;
17 }
```

Output:

```
Input numbers of terms:10
The odd numbers are:1 3 5 7 9 11 13 15 17 19
The Sum of odd Natural up to 10 terms:100

Process returned 0 (0x0)   execution time : 2.952 s
Press any key to continue.
|
```